

Scenario for CS2043 Submission #4 (Group 13) – Version 1 & Version 2

Here is the test data structure for the scenario's (which can also be downloaded from the github zip file.)

CS2043_TestDataV2/

Cases/

Programs/

Results/

The Programs folder contains subfolders for students and their .java files.

The Cases folder contains text-based test case definitions.

The Results folder is where Version 2 result files will be stored.

Version 1 Scenario

(Create → Load → Add to Suite → Execute Suite → Display Results)

1. Run the UI Class (JavaFX)

- The main UI window appears.

NEXT

2. Create a Test Case (Manual)

- Click “**Create Test Case (manual)**”.
- A popup opens with:
 - **Title**
 - **Input**
 - **Expected Output**

Now we can create our title, input and expected output...

- Title: **AddTwoNumbers**
- Input: 3 4
- Expected Output: 7
- Press **Enter**.

NEXT

3. Load a Test Case from File (if you want to load your own test case)

- Click “**Load Test Case from file**”.
- Popup appears with a text field for the filename and a bold label: “**Enter full path to test case file above (without quotations).**”

Example:

- Enter:
CS2043_TestDataV2/Cases/T2.txt
- Press **Enter**.

- If the file follows the required structure
(Line 0 = title, Line 1 = input, Line 2 = expected output),
Coordinator loads the test case and you shouldn't have any issues.

4. Add Test Case to Suite (Only 1 test suite so if you want to be able to execute the test suit, you have to add all the test cases you want to run to the suite)

- Click “**Add Test Case to Suite**”.

Example:

- Enter: AddTwoNumbers (Title)
- Press **Enter**.
- Status updates:
“**Added AddTwoNumbers (TestCase) to Test Suite.**”

Version 1 supports only **one single Test Suite**, so all test cases go into the same suite.

5. Execute Test Suite

- Click “**Execute Test Suite**”.
- A popup appears with:
 - An **Enter** button
 - A **Save Results** button

Required Before Running:

The user must set the root folder using the Version 2 button:

Set Root Folder → CS2043_TestDataV2/Programs

Running:

- Click **Enter**.
- The Coordinator:

- populates all student programs from subfolders
e.g.,
- Programs/StudentA/StudentA.java
- Programs/StudentB/StudentB.java
- compiles each program
- runs each test case
- compares program output vs expected results

Example Result Display:

Program: StudentA

TestCase "AddTwoNumbers" -> output: 7 (correct)

Program: StudentB

TestCase "AddTwoNumbers" -> output: 12 (wrong)

6. Displaying the Result

This output is visible inside the Execute TS popup and also stored internally for Version 2 saving.

This ends the Version 1 scenario.

Version 2 Scenario

(Folder Selection → Saving Results → Loading Results → Comparing Results)

Version 2 adds additional UI features and result management.

7. Set Root Folder (Version 2 Feature)

- From the main UI, click **Set Root Folder**.

Example:

CS2043_TestDataV2/Programs

- Click **Enter**.
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8. Set Results Folder (Version 2 Feature)

- From the main UI, click **Set Results Folder**

Example:

CS2043_TestDataV2/Results

- Press **Enter**.

9. Save Results After Executing Test Suite

- Open **Execute Test Suite**, click **Enter** to run.
- After the result output appears, a **Save Results** button becomes visible.
- Click **Save Results**.

What happens:

- The system writes the entire result text to a file in the Results folder.

- File naming format:

results_<rootFolderName>_<timestamp>.ser

Example saved file:

CS2043_TestDataV2/Results/results_Programs_1733749821450.ser

- Status updates:
 **“Results saved to:
 CS2043_TestDataV2/Results/results_Programs_1733749821450.ser”**
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10. Load Results

- Click **Load Results** on the main UI.
- A popup appears listing all .ser result files from the Results folder.
- Bold label:
 “Select a results file below and click Enter to view it.”
- Select a file → Press **Enter**.

Example:

Selected file:

results_Programs_1733749821450.ser

Display:

The content of that file is shown in the text area (the same detailed textual results produced by Execute TS).

Status updates:

“Loaded results file: results_Programs_1733749821450.ser”

11. Compare Result Files

- Click **Compare Result Files**.
- A popup opens listing all .ser results, each with a checkbox.
- Bold label:
“Select exactly TWO results files using the checkboxes below, then click Enter to compare.”

Steps:

1. Select exactly **two** result files.
2. Press **Enter**.

The system:

- Reads both files.
- Displays the contents one after the other for human comparison.

Example Comparison Display:

(Input was 3 and 4)

===== results_Programs_1733749821450.ser =====

Program StudentA: TestCase “AddTwoNumbers” -> output: 7

Program StudentB: TestCase “AddTwoNumbers” -> output: 12 (wrong)

===== results_Programs_1733750129910.ser =====

Program StudentA: TestCase “AddTwoNumbers” -> output: 7

Program StudentB: TestCase “AddTwoNumbers” -> output: 10 (wrong)

Status updates:

“Compared result files: file1 and file2”
